Uttapreksha Patel

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Engineer who owns problems end to end, from ambiguous requirements through deployed systems: built a full-stack AI product solo, integrated LLM APIs and retrieval systems, and debugs production issues down to the root cause. Strong Python/TypeScript fundamentals, writes technical work up clearly for mixed audiences.

Technical Skills

Full-Stack	React, TypeScript, FastAPI, PostgreSQL, REST APIs
AI Integration	LLM APIs, RAG, Embeddings, Agentic Workflows
Languages	Python (OOP, PEP8), TypeScript, C++, SQL
Infra	Docker, Kubernetes, AWS EC2, CI/CD
Writing	Technical Writing, Public Presentations, Documentation

Work Experience

Northeastern University | *Collaboration with Sandia National Laboratories* — **Boston, MA**
Research Assistant | Systems Engineering & Ambiguous Problem-Solving — *February 2025 – Present*

- Scoped and shipped Python data pipelines end-to-end without a fully specified brief, processing **1M+ operation records** with automated anomaly detection; deployed to production on AWS EC2, improving pipeline reliability **30%**
- Translated real-world constraints into elegant engineering solutions: designed controlled experiments isolating exactly which resource caused a performance regression, quantifying effects with sub-second precision across **8 identified events**
- Deployed telemetry infrastructure and diagnosed critical buffer-overflow failures through memory allocation tuning, restoring stable service under multi-node production load

TecUnique Pvt. Ltd. — **Vadodara, India**
Python Development Intern — *Dec 2023 – Apr 2024*

- Built Python REST API automation framework using Pytest achieving **93% pass rate across 30 test cases**; automated end-to-end validation with structured HTML reports and regression tracking
- Optimized SQL queries and numerical pipelines, reducing runtime **20%**; managed source control through Git in an Agile/Scrum environment

Relevant Projects

Clinical Intake Agent: Full-Stack AI Product, Owned Solo End to End | *Python, FastAPI, Celery, Redis, React, TypeScript, Claude API, Llama, PostgreSQL, Docker, GitHub Actions | GitHub | Blog*

- Owned a 6-step production system end-to-end, solo, from architecture through deployment: React/TypeScript frontend, FastAPI backend orchestrated via async Celery workers, PostgreSQL schema design, containerized with CI on every commit
- Used AI tools (Claude) extensively, applying critical judgment rather than accepting outputs blindly: built a golden-set eval harness with LLM-as-judge graders; the harness reported **90.6% extraction / 0% hallucination** and passed, but manual review still caught a real failure it missed, driving a concrete fix

Healthcare Knowledge Retrieval API: Integrations, Production Deployment | *Python, FastAPI, FAISS, PostgreSQL, Docker, AWS EC2, Claude API, OpenAI API*

- Built and integrated a backend system over **50K+ unstructured documents**: data ingestion, LLM API integrations, FAISS vector search, and PostgreSQL-backed relational storage, designed and implemented end-to-end
- Iterated through systematic evaluation to improve system correctness **+25%** and cut latency **-22%**; containerized with full integration test coverage and continuous production monitoring

Full-Loop LLM Agent: Agentic Workflows, Deployed End to End | *Python, Claude API, OpenAI API, LangGraph, RAG, FAISS, FastAPI, PostgreSQL, Docker | GitHub*

- Built a LangGraph-orchestrated agent with tool use and adaptive memory, applying guardrails and fallback logic to recover from failures without human intervention across 75 dialogues and 15 domains
- Built an evaluation framework from scratch, LLM-as-judge graders and a 4-category failure-mode taxonomy benchmarked against 4 baselines, improving task accuracy **0.613** $\rightarrow$ **0.760** and cutting failures **64%**

PitchQuest: Multi-Model Integrations, Presented to External Audiences | *PyTorch, Whisper, MediaPipe, DeepFace, SmolVLM, LoRA, FastAPI, Weights & Biases | GitHub*

- Built an end-to-end system integrating **5 foundation models** in parallel across text, voice, and vision, with a FastAPI backend tying the integrations together
- Achieved **85%+ model agreement** with end-to-end latency under 2 minutes; presented the work live to external audiences at Google and Cambridge

Research

Analyzing VLM-Based Approaches for Anomaly Classification and Segmentation | *Co-authored, arXiv 2026*

- Evaluated WinCLIP and AnomalyCLIP on MVTec AD and VisA benchmarks; achieved **0.916 classification AUROC** and **0.907 segmentation AUROC** through reproducible evaluation pipelines and systematic error analysis

Education

Northeastern University — **Boston, MA**
M.S. Electrical and Computer Engineering (GPA: 3.9/4.0) — *May 2026*

G H Patel College of Engineering and Technology — **Anand, India**
B.E. Electronics and Communication Engineering (GPA: 3.7/4.0) — *May 2024*